

BASIC ELECTRONICS

(xxx)

UNIT-2

Lecture-5&6 (Full Wave Rectifier)

Topics to be Discussed

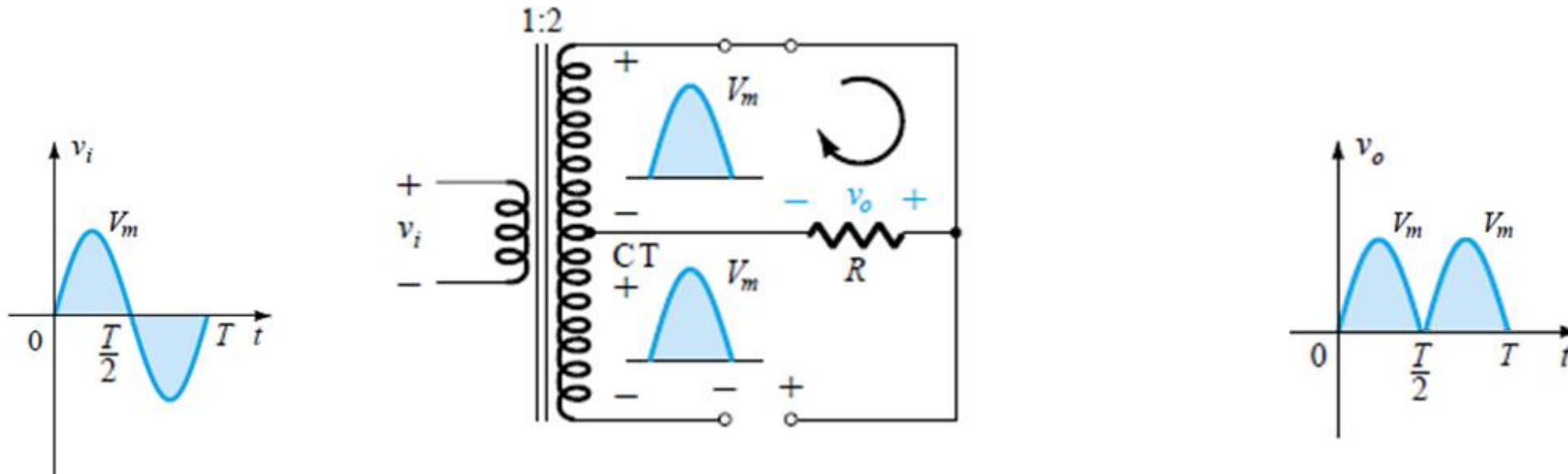
- **Full Wave Rectifier**
- **Center Tapped Full Wave Rectifier**
- Parameters of the Full Wave Rectifier
- Efficiency of Full Wave Rectifier
- Ripple Factor
- Peak Inverse Voltage
- Peak Factor and Form Factor
- Effect of Using a Silicon diode with $V_T = 0.7V$
- Examples
- **Full Wave Bridge Rectifier**
- Peak Inverse Voltage
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- Advantages and Disadvantages of Full Wave Bridge Rectifier
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Full Wave Rectifier [1, 2]

- The full wave rectifier is a very useful circuit for **AC to DC conversion** and it is designed using the diodes.
- Unlike the half wave rectifier, the full wave rectifier gives **output during both half cycles**.
- The output of the Full wave rectifier is **pulsating DC**.
- The ripple in the output waveform can be reduced using the filter.
- **There are two types of full wave rectifier circuits as follows:**
 - **Center tapped full wave rectifier**
 - **Full wave bridge rectifier**

Center Tapped Full Wave Rectifier [1]

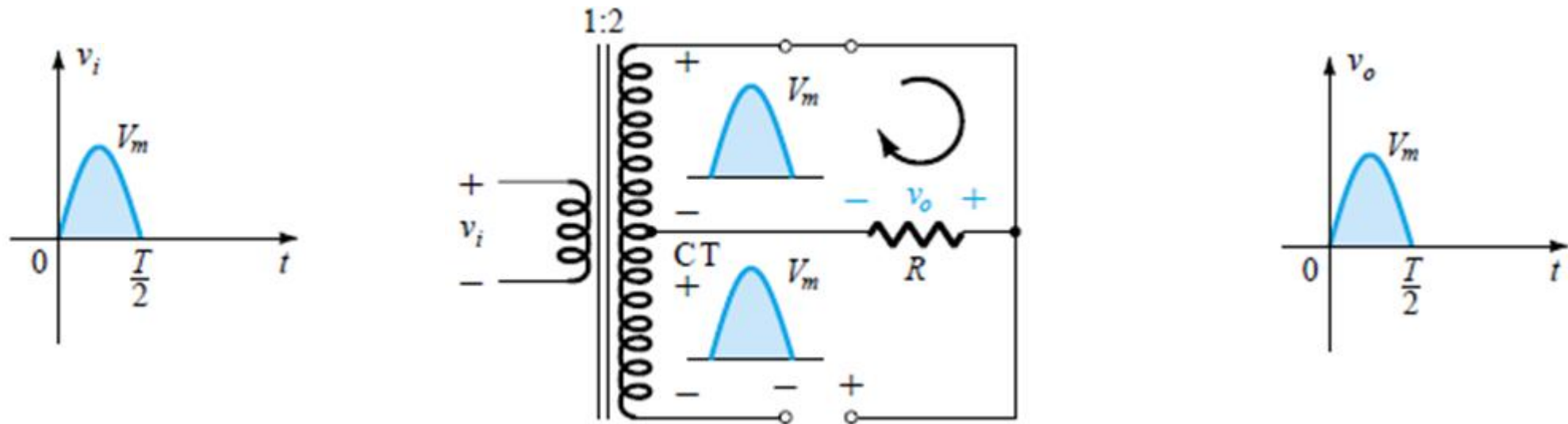
- A **popular** full wave rectifier appears in following Figure with **only two diodes** but requiring a **center-tapped (CT) transformer** to establish the input signal across each section of the secondary of the transformer.



Center Tapped Full Wave Rectifier [1]

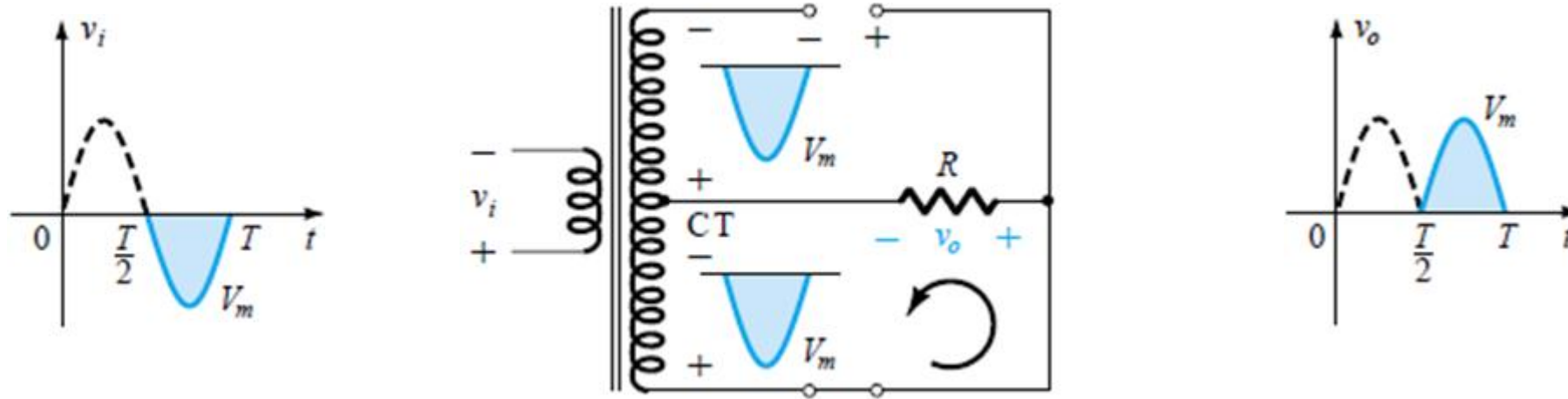
Operation:

- **Case 1:** During the **positive portion** of ' v_i ' applied to the primary of the transformer, the network will appear as shown in Figure.



Center Tapped Full Wave Rectifier [1]

- **Case: 2** During the **negative portion of the input**, the network appears as shown in Figure.



- Reversing the roles of the diodes but maintaining the same polarity for the voltage across the load resistor, R .
- The **net effect** is the same output as that appearing in **positive half cycle** with the same **DC levels**.

Parameters of the Full Wave Rectifier [1, 2]

- Let $v_i = V_m \sin \omega t$ be the input voltage to the rectifier, where V_m is the peak input voltage.
- Let r_f is the resistance in the forward direction i.e. in the 'ON' state, and $R_R (= \infty)$ in the reverse direction i.e. in the 'OFF' state.

$$i_1 = \begin{cases} I_m \sin \omega t & \text{for } 0 \leq \omega t < \pi \\ 0 & \text{for } \pi \leq \omega t \leq 2\pi \end{cases}$$

$$i_2 = \begin{cases} 0 & \text{for } 0 \leq \omega t < \pi \\ I_m \sin \omega t & \text{for } \pi \leq \omega t \leq 2\pi \end{cases}$$

where,

$$I_m = \frac{V_m - V_T}{r_f + R_L}$$

Average Current (I_{dc}):

$$I_{dc} = \frac{1}{\pi} \int_0^{\pi} i_1 d\omega t$$

$$= \frac{1}{\pi} \left(\int_0^{\pi} I_m \sin \omega t d\omega t + 0 \right)$$

$$= \frac{2I_m}{\pi} = 0.636 I_m$$

or $I_{dc} = \frac{2(V_m - V_T)}{\pi(r_f + R_L)}$

If $R_L \gg r_f$, and $V_T = 0$ then, $I_{dc} = \frac{2V_m}{\pi R_L}$

Average Voltage (V_{dc}):

$$V_{dc} = I_{dc} \times R_L = \frac{2I_m}{\pi} \times R_L = \frac{2(V_m - V_T)}{\pi(r_f + R_L)} \times R_L = \frac{2(V_m - V_T)}{\pi \left(1 + \frac{r_f}{R_L} \right)}$$

If $R_L \gg r_f$, and $V_T = 0$ then $V_{dc} = \frac{2V_m}{\pi} = 0.636 V_m$

RMS Current (I_{rms}): Since current is of the same form in the two halves.

$$\begin{aligned} I_{rms} &= \sqrt{\frac{1}{\pi} \int_0^{\pi} i^2 d\omega t} = \sqrt{\frac{1}{\pi} \left[\int_0^{\pi} I_m^2 \sin^2 \omega t d\omega t \right]} \\ &= \sqrt{\frac{I_m^2}{\pi} \int_0^{\pi} \frac{(1 + \cos 2\omega t)}{2} d\omega t} = \frac{I_m}{\sqrt{2}} \end{aligned}$$

RMS Voltage (V_{rms}):

$$V_{rms} = I_{rms} \times R_L = \frac{I_m}{\sqrt{2}} \times R_L \times \frac{V_m - V_T}{\sqrt{2}(r_f + R_L)} \times R_L = \frac{V_m - V_T}{\sqrt{2} \left(1 + \frac{r_f}{R_L} \right)}$$

If $R_L \gg r_f$, and $V_T = 0$ then

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

Efficiency of Full Wave Rectifier [1, 2]

- The ratio of DC power output to the applied input AC power is known as rectifier efficiency.

$$\text{Rectifier efficiency, } \eta = \frac{\text{DC power output}}{\text{Input AC power}} = \frac{P_{dc}}{P_{ac}}$$

$$\frac{P_{dc}}{P_{ac}} = \frac{I_{dc}^2 \times R_L}{I_{rms}^2 \times (R_L + r_f)} = \frac{\left(\frac{2I_m}{\pi}\right)^2 \times R_L}{\left(\frac{I_m}{\sqrt{2}}\right)^2 \times (R_L + r_f)} = \frac{8}{\pi^2} \frac{R_L}{(R_L + r_f)}$$

$$\eta = \frac{0.812}{\left(1 + \frac{r_f}{R_L}\right)}$$

- The efficiency of a center tapped full wave rectifier is equal to 81.2% (i.e. $\eta_{\max} = 81.2\%$)

Ripple Factor [1, 2]

- Ripple Factor is the ratio of RMS value of ac component present in the rectified output to the average value of rectified output. It is a dimensionless quantity and denoted by ' r '.

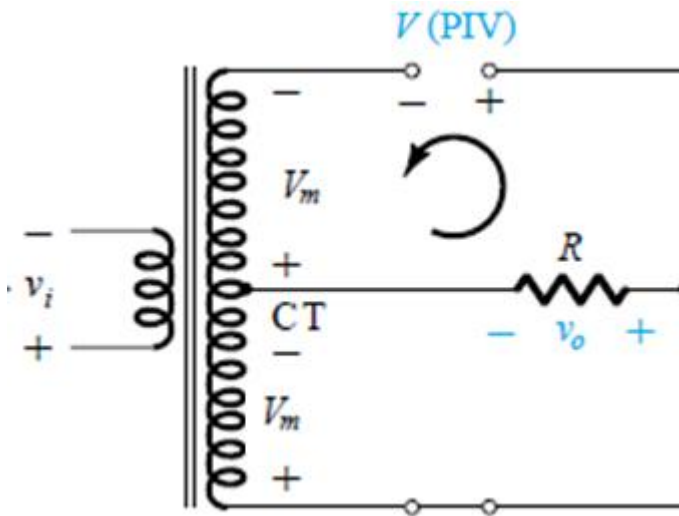
$$\text{Ripple factor } (r) = \sqrt{\frac{I_{\text{rms}}^2 - I_{\text{dc}}^2}{I_{\text{dc}}^2}} = \sqrt{\frac{V_{\text{rms}}^2 - V_{\text{dc}}^2}{V_{\text{dc}}^2}}$$

Ripple factor of full wave rectifier:

$$\text{Ripple factor } (r) = \sqrt{\frac{\left(\frac{I_m}{\sqrt{2}}\right)^2}{\left(\frac{2I_m}{\pi}\right)^2}} - 1 = 0.48$$

Peak Inverse Voltage [1, 2]

- Peak inverse voltage (PIV) or peak reverse voltage (PRV) is the **maximum voltage that the diode can withstand without breakdown during reverse bias condition.**
- The required PIV rating for the Center tapped full wave rectifier can be determined as shown below.



- Applying Kirchhoff's voltage law,

$$V_{PIV} = V_m + v_o = V_m + V_m = V_{PIV} \geq 2V_m$$

Higher PIV rating compared to a **bridge rectifier** (where $PIV = V_m$ only)

Peak Factor and Form Factor [1, 2]

- For the full wave rectifier, the expressions for peak factor and form factor are as follows:

Peak Factor:

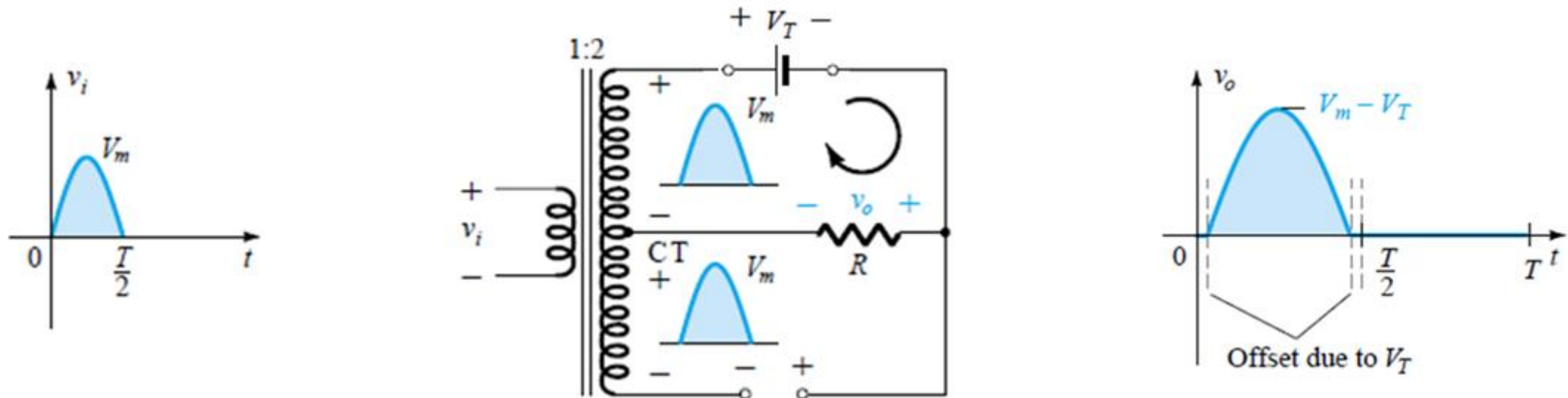
$$\text{Peak Factor} = \frac{I_m}{I_{r.m.s}} = \frac{I_m}{\frac{I_m}{\sqrt{2}}} = \sqrt{2} = 1.414$$

Form Factor:

$$\text{Form Factor} = \frac{I_{r.m.s}}{I_{avg}} = \frac{\frac{I_m}{\sqrt{2}}}{\frac{2I_m}{\pi}} = \frac{\pi}{2\sqrt{2}} = 1.11$$

Effect of Using a Silicon Diode with $V_T = 0.7V$ [1]

- For the forward-bias region in the FWR:



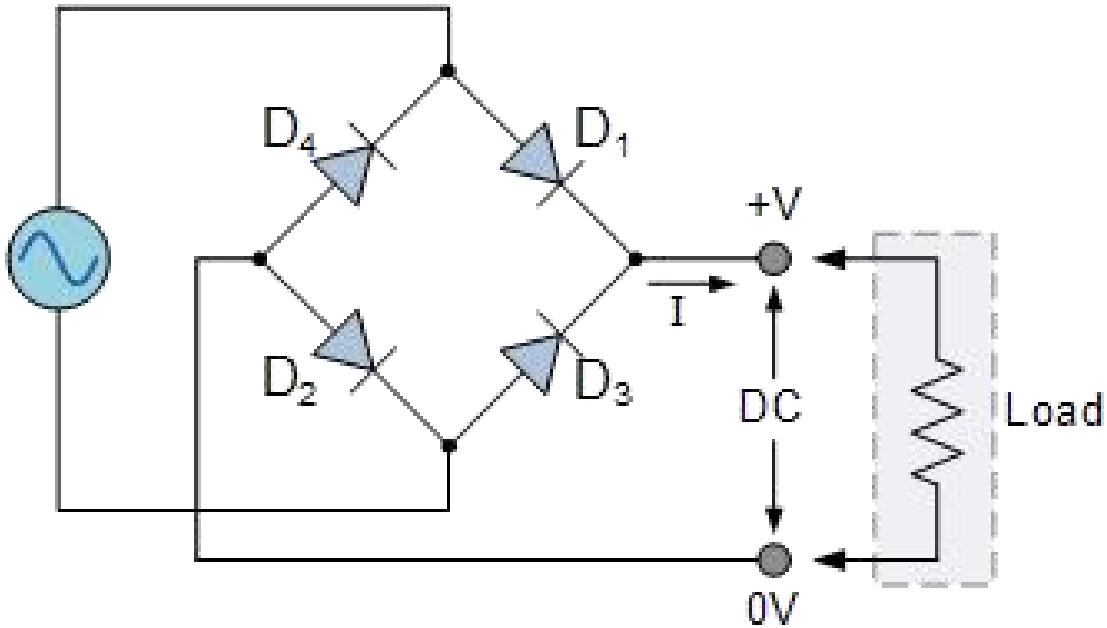
- The **applied signal** must now be **at least $V_T = 0.7V$** before the **diode can turn 'ON'**.
- When conducting, the difference between v_o and v_i is a fixed level of $V_T = 0.7V$ and $v_o = v_i - V_T$.

$$V_{dc} \cong 0.636(V_m - V_T)$$

Properties	Center Tapped Full wave rectifier	
	Ideal diode	Diode with V_T
D.C. Current (I_{dc})	$2I_m/\pi$	$2I_m/\pi$
Maximum Current (I_m)	$V_m/(r_f+R_L)$	$(V_m-V_T)/(r_f+R_L)$
D. C. voltage (V_{dc})	$0.636V_m$	$0.636(V_m-V_T)$
Maximum efficiency	$0.812/(1+(r_f/R_L))$	$0.812/(1+(r_f/R_L))$
Peak Inverse Voltage	$2V_m$	$2V_m-V_T$

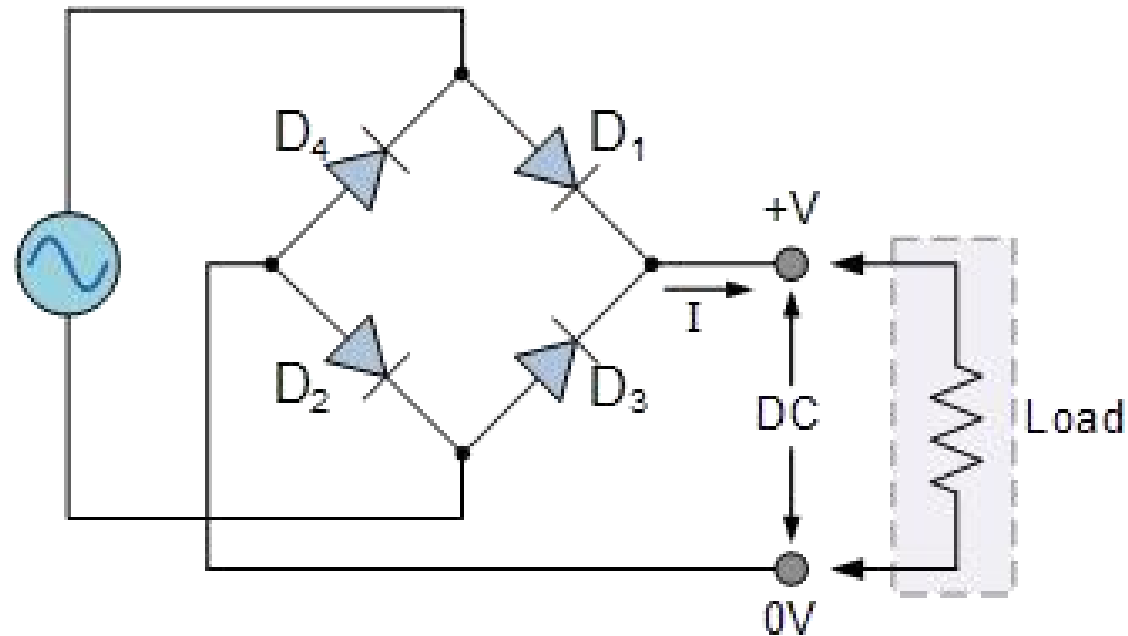
Full Wave Bridge Rectifier [1, 2]

- Another type of circuit that produces the same output waveform as the full wave rectifier circuit, is the ***Full Wave Bridge Rectifier***.
- The main advantage of this bridge circuit is that it **does not require a special center tapped transformer**, thereby **reducing its size and cost**.



Full Wave Bridge Rectifier [1, 2]

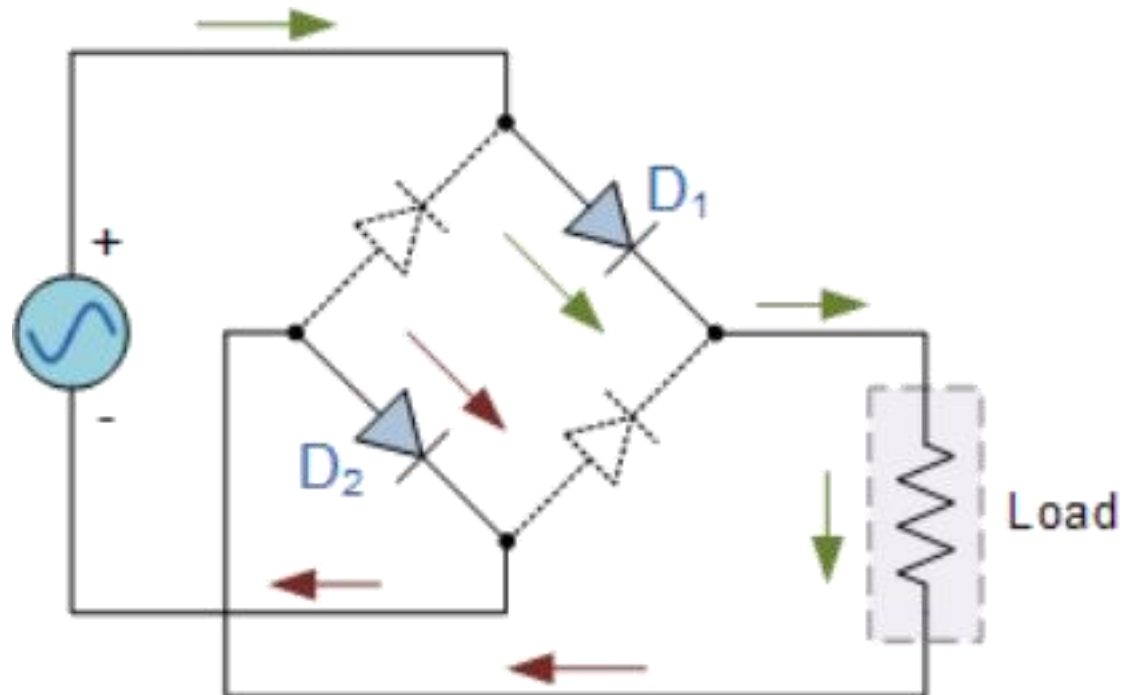
- The four diodes labelled D_1 to D_4 are arranged in “series pairs” with only two diodes conducting current during each half cycle.



Full Wave Bridge Rectifier

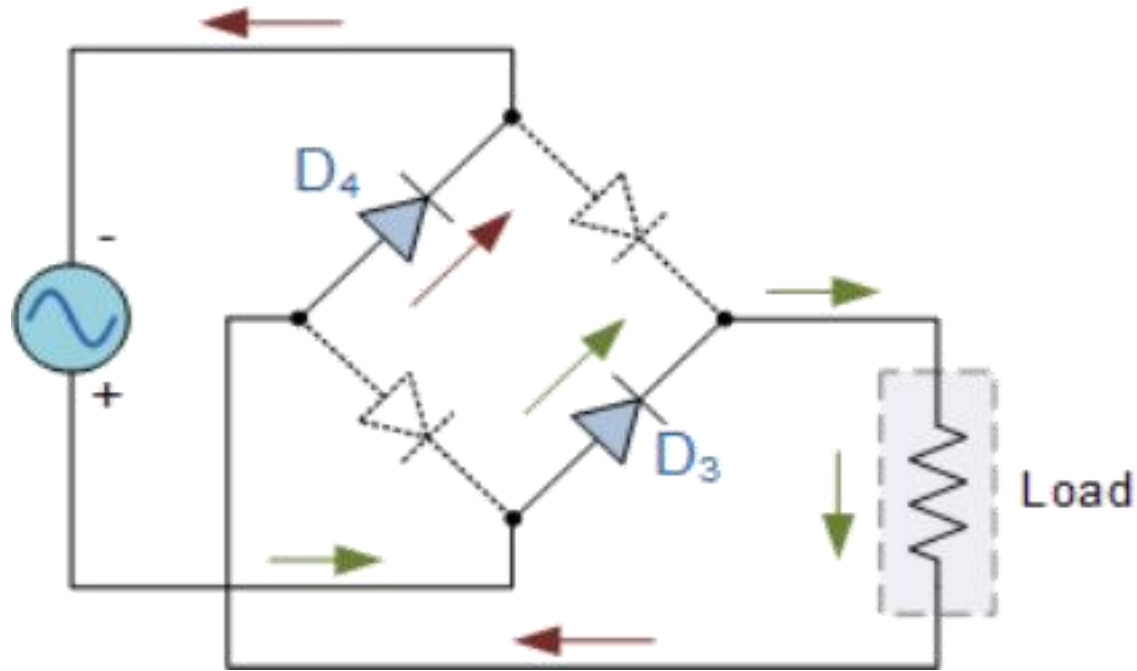
Operation:

- **Case 1:** During the positive half cycle of the supply, diodes D_1 and D_2 conduct in series while diodes D_3 and D_4 are reverse biased.



Full Wave Bridge Rectifier

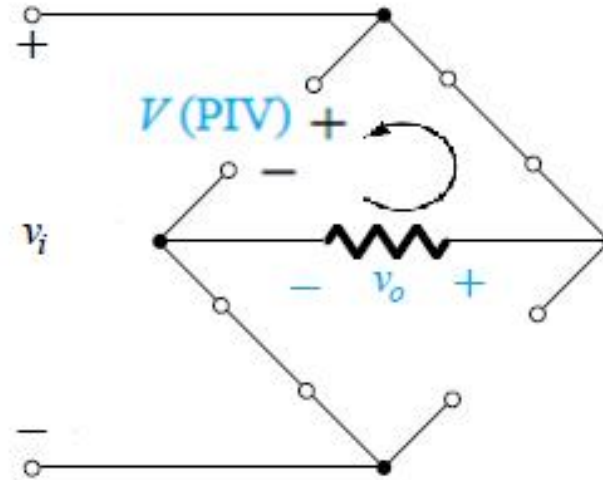
- **Case 2:** During the negative half cycle of the supply, diodes D_3 and D_4 conduct in series, but diodes D_1 and D_2 switch “OFF” as they are now reverse biased.



Note: The current flowing through the load is the same direction as before in the positive half cycle.

Peak Inverse Voltage [1,2]

- The required PIV rating for the full-wave bridge rectifier can be determined as shown below.



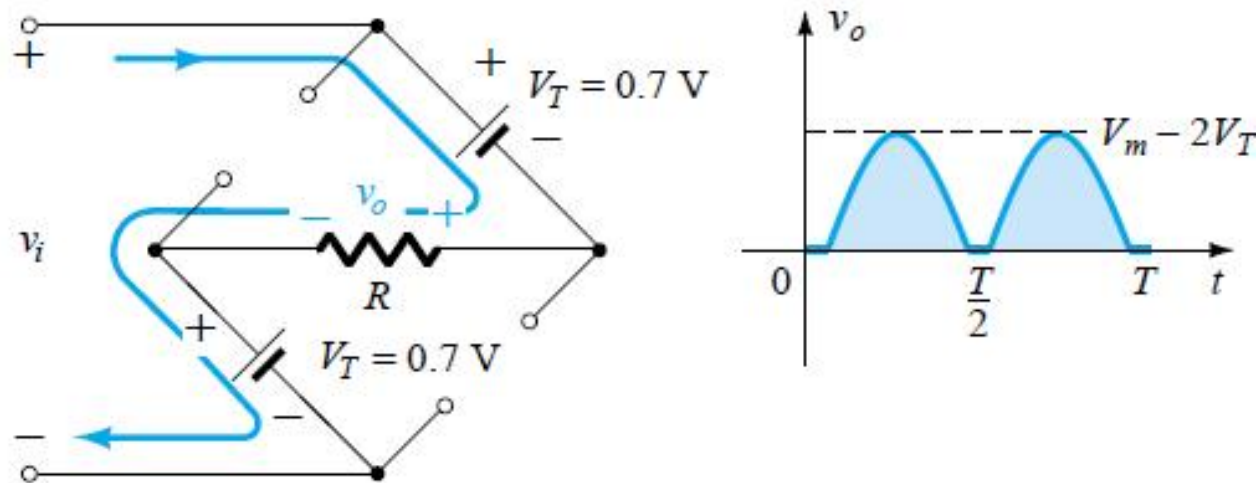
- Applying Kirchhoff's voltage law,

$$V_{PIV} = v_o = V_m$$

$$V_{PIV} \geq V_m$$

Effect of Using a Silicon Diode with $V_T = 0.7\text{V}$ [1]

- For the forward-bias region in the FWR:



- The applied signal must now be at least $V_T = 0.7\text{V}$ before the diode can turn 'ON'.
- When conducting, the difference between v_o and v_i is a fixed level of $V_T = 0.7\text{V}$ and $v_o = v_i - 2V_T$.

$$V_{dc} \cong 0.636(V_m - 2V_T)$$

Properties	Full Wave Bridge Rectifier	
	Ideal diode	Diode with V_T
D.C. Current (I_{dc})	$2I_m/\pi$	$2I_m/\pi$
Maximum Current (I_m)	$V_m/(2r_f+R_L)$	$(V_m-2V_T)/(2r_f+R_L)$
D. C. voltage (V_{dc})	$I_{dc}R_L$	$I_{dc}R_L$
Maximum efficiency	0.812	$0.812/(1+(2r_f/R_L))$
Peak Inverse Voltage	V_m	V_m-2V_T

Advantages and Disadvantages of Full Wave Bridge Rectifier

Advantages:

- Need for center tap transformer is eliminated.
- PIV is one half of that of center tap circuit.

Disadvantages:

- Requires 4 diodes.
- Internal resistance voltage drop is twice than that of Center Tap Circuit.

References

- [1] R. Boylestad and L. Nashelsky, 'Electronic Devices and Circuit Theory', PHI, 7e, 2001.
- [2] D.C. Kulshreshtha, 'Electronic Devices and Circuits', New Age, 2e, 2006.